

Analysis of Network on Chip Topologies

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Abstract—The incorporation of multi-core processors using current technologies is a challenging task as establishing effective communication among distant cores consumes significant time, ultimately diminishing the benefits of multi-core functionality. A new technology called Network on Chip (NoC) has been introduced to address this issue and is considered the future of the semiconductor industry. To develop new multi-core processors that outperform existing processors, it is crucial to compare traditional technologies with NoC. The traditional method faces difficulties implementing multi-core systems effectively and establishing efficient communication among different logic units, making NoC a helpful tool for developing multi-core processors. A detailed study of the NoC architecture is necessary, including the understanding of each component of its basic building blocks, such as topology, flow control mechanisms, routing logic, and router micro-architecture, and their performance comparison. This understanding is critical to select the most appropriate NoC architecture for implementing multi-core processors.

Keywords—Network on chip, Network-on-chip topology, Routing Algorithms, Router Architecture.

I. INTRODUCTION

Single bus architecture refers to a computer system design where all components in the system are connected to a single bus. The bus is a shared communication pathway that allows the components to exchange data and instructions with each other. In single bus architecture, all data transfers are made through the same bus, which can create a bottleneck in system performance. This is because the bus must be shared among multiple components, and only one component can access the bus at a time. Therefore, if multiple components are trying to access the bus simultaneously, they must wait for their turn, which can slow down the system's overall performance. This single bus architecture is not adaptable in multi-core system because of its inefficiency to establish the communication among multiple logic units. Figure 1 shows the schematic diagram of single bus architecture.

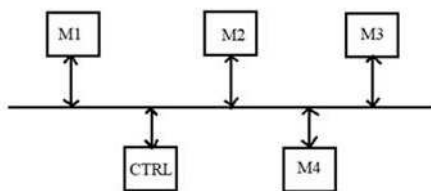


Fig. 1. Single bus architecture.

The introduction of multi bus architecture was aimed at addressing the performance limitations inherent in single bus architecture. The multi-bus architecture facilitates communication among multiple logic units by providing more than one bus for communication. This allows more than a pair of units to communicate simultaneously based on the availability of the bus, resulting in improved communication

efficiency. The architecture typically supports bus connections between 3 to 10 communication partners effectively [1]. However, it may not be suitable for larger numbers of communication partners. Figure 2 illustrates the multi-bus system architecture. The rapid enhancement in the VLSI technology and lithography process allowed more and more logics to be implanted on silicon chips. Whenever the number of modules/logics increases the performance of the multi bus architecture degrades. This led to slow down the entire process and causes the performance degradation of chips. So, to overcome the issue in performance and availability of silicon area system on chip is introduced.

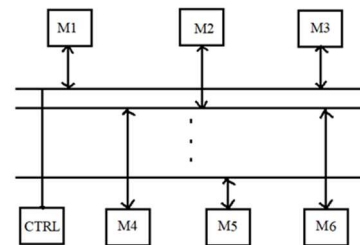


Fig. 2. Multibus architecture.

A System on Chip (SoC) is an integrated circuit (IC) that integrates all the components and functionalities of a computer or electronic system onto a single chip. This includes digital and analog circuits, memory, processors, input/output interfaces, and other necessary components for a particular application or system. SoC has many benefits over traditional computing systems that use separate chips for different functions.

A bus-based system on chip (SoC) is a type of integrated circuit that uses a shared communication bus to connect the various components on the chip. In a bus-based SoC, there is a single communication pathway that is used by all the components on the chip to exchange data and control signals. The bus is typically controlled by a central bus controller that manages the flow of data and arbitrates access to the bus by different components. The main advantage of a bus-based SoC is its simplicity and low cost, as there is only one communication pathway to manage. However, as the number of components on the chip increases, the bus can become a bottleneck, limiting the performance of the system. Even though we could implant more silicon on chips, common bus architecture was a major problem in terms of communication. This led to the implanting of a new communication technology, which is called network on chip.

Network on chip (NoC)[2][3] is a communication architecture for designing complex integrated circuits (ICs). It is a packet-based communication network that uses a set of interconnected routers to provide scalable, flexible, and efficient communication between different components and subsystems of the chip.

In NoC, data is transmitted in the form of packets, which are routed through a network of switches or routers. Each router has its own address, and packets are sent from one router to another based on their destination address. The routers work together to ensure that the packets arrive at their destination in a timely and efficient manner. Figure 3 depicts the internal view of NoC[2][4]. In the internal structure of a Network on Chip (NoC), the chip is divided into smaller units called tiles, each containing one or more processing cores, which can be any programmable units responsible for executing computations or tasks. These tiles are interconnected by routers, which manage the routing of data packets between the cores, allowing for efficient communication and data transfer within the network.

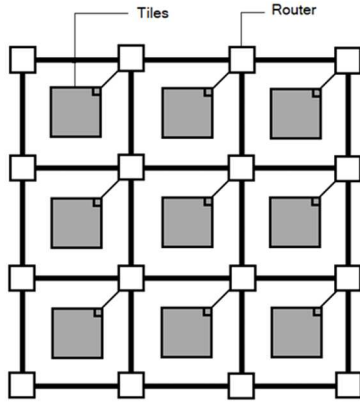


Fig. 3. Internal view of network on chip.

NoC-based SoC can provide higher performance and scalability, as the network can be designed to accommodate a larger number of components and provide more efficient communication between them. NoCs have several advantages over traditional bus-based communication architectures. They are more scalable, as they can handle a larger number of components and provide better bandwidth utilization.

The upcoming section provides an in-depth exploration of network on chip architecture, encompassing a range of topologies and flow control mechanisms. It delves into the internal architecture of NoC, addresses significant issues associated with different topologies, and presents an analysis of the advantages and disadvantages of various flow control mechanisms.

II. RELATED WORKS

A. Network on Chip Architecture

The NoC architecture replaces traditional bus architecture with a network. Communication is established in the form of packets transferred through the network. Typically, a modular NoC architecture is comprised of three essential components: Network Interface (NI), Switches and Links.

The Network Interfaces' primary function is to convert the bus protocol used by the Processing Elements (PEs) to the network protocol used by the switches. Each IP core must be connected to the NoC using an NI [5][6]. Transaction requests and responses are converted into packets by NIs, and vice versa. The packets are serialized into flits by the NI. The core of the network is made up of switches. Routing packets from source to destination is their primary duty. In order to provide connectivity for many PE, they enable the creation of various topologies and provide arbitrary connectivity between

numerous inputs and outputs. Buffering resources are made available by switches to reduce congestion and enhance performance. The connectivity between NIs and switches, as well as between the switches themselves is abstracted via links. Links are more than just actual wires because they can act as a pipe to accomplish the required timing. Figure 4 shows the internal architecture of NoC. In this PE indicates Processing Elements, NI indicates Network Interface and RL indicates the Routing Logic.

The network contains routers and wires; each router consists of processors, memories and other blocks. The implementation of the NoC must meet the performance requirements which are guaranteed throughput, low latency, diversity in routing, sufficient data transfer capacity and consuming less power.

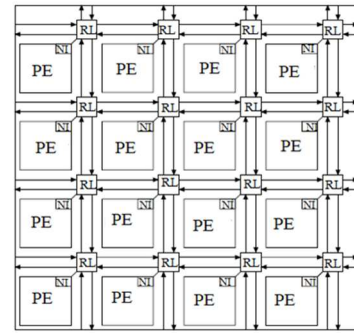


Fig. 4. Internal architecture of network on chip

The architectural requirements considered in the implementation of NoC are scalability, programmability and generality. Besides these, NoC implementations provide the benefits such as modularity, NoCs can be tuned to support specific applications on SoC, and enhances reliability. Topology, routing logic, flow control, and router micro architecture are the pillars of Network on Chip [7][8].

Generally, two kinds of network traffic are preferred for NoC. They are Guaranteed Throughput (GT)[2] traffic and Best Effort (BE) traffic. GT traffic performs best with routing algorithms, which act like circuit-switched networks. The BE traffic sending packets never guarantees that the packets will reach the receiver. The packet-switched networks are mostly BE traffic [7][9].

B. Network on Chip Topologies

Topologies are used for interconnecting different modules in NoC. Using a routing method, the packets are directed to the various modules according to the topology. The most commonly used topologies are; mesh, torus, tree, butterfly, polygon, star and Ring Connected Binary Tree (RiCoBiT)[10][11][12].

1) Mesh Topology

In the mesh topology [9][13], the interconnection has m rows and n columns and is shaped like a matrix. At the intersection of the row and columns are the routing devices and other computational units. The X and Y coordinates can be used to identify the modules. Figure 5 shows the mesh topology. In a Mesh topology, the latency for communication between nodes at opposite ends of the network can be higher than in other topologies.

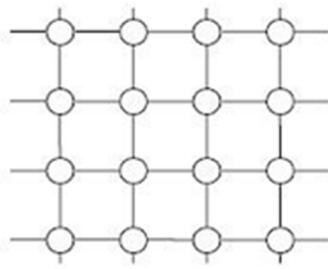


Fig. 5. Mesh topology.

2) Torus Topology

A Torus network [9][12][13] is a type of enhanced mesh network. A basic torus network is a mesh in which the heads and tails of the columns and the left and right sides of the rows are linked. In comparison to the mesh network, the torus network offers significantly more paths and more minimum routes. The topology of a torus is depicted in Figure 6. The torus network is complex to design and it provides higher latency for long distance communication.

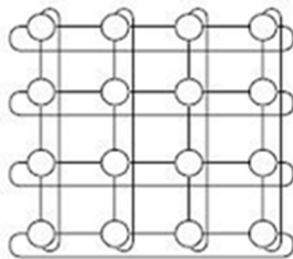


Fig. 6. Torus topology.

3) Tree Topology

In tree architecture [7][9][13] the nodes serve as routers, while the leaves function as computational resources. A leaf node's ancestors are the routers located above it, and its descendants are the leaf nodes situated beneath the ancestor. In a fat tree topology, each node has redundant predecessors, resulting in numerous alternative routes that can link nodes together. Figure 7 displays the tree topology. The scalability and flexibility is limited in tree topology.

4) Butterfly Topology

A butterfly network is a network that has a butterfly-shaped structure and usually uses deterministic routing. This type of network can be either unidirectional or bidirectional. In a simple unidirectional butterfly network, there are 8 input ports, 8 output ports, and 3 levels of routers, each containing 4 routers. The packets entering from the left side of the network are routed to their respective output ports on the right side[5][9]. In a bidirectional butterfly network, all inputs and outputs are on the same side of the network. The packets arriving at the inputs are first routed to the other side of the network, then turned around and routed back to the correct output. A butterfly network consists of two stages of routers, each stage containing two routers, with four inputs and four outputs is shown in Figure 8. One major drawback of using a butterfly topology in a Network-on-Chip (NoC) is that it can be less scalable and more complex than other topologies, especially as the number of nodes in the network increases.

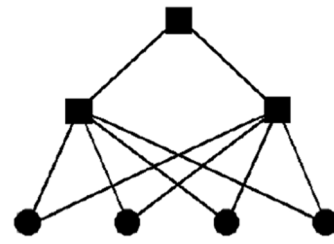


Fig. 7. Tree topology.

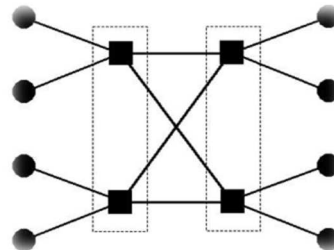


Fig. 8. Butterfly topology.

5) Polygon Topology

The circular polygon network[9][13] is the most commonly used type of network for transmitting packets between routers. The addition of chords to the circle increases the network's versatility. Figure 9 illustrates the polygon topology, which is referred to as the spidergon topology when the chords are only present between opposing routers. One major drawback of using a polygon topology in a Network-on-Chip (NoC) is that it can lead to routing congestion and increased latency.

6) Star Topology

A star network [8][9][13] consists of a central router at the hub, connected to various computing power or smaller networks at the spikes. However, since the central router is responsible for managing all the communication between the spikes, it requires a high capacity, which often leads to congestion in the core of the star network. The Figure 10 illustrates the configuration of the star topology. One of the major drawbacks of the star topology in NoC is that it relies heavily on a central hub or router, which can become a bottleneck if there is a large amount of traffic flowing through the network.

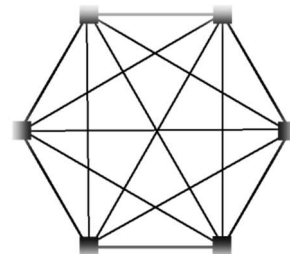


Fig. 9. Polygon topology.

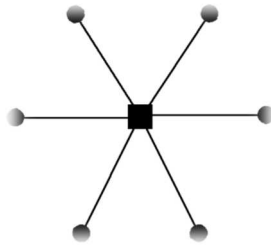


Fig. 10. Star topology.

7) Ring Connected Binary Tree (RiCoBiT)

RiCoBiT[10][11][12] is created by joining a binary tree with expanding concentric rings, shown in the Figure 11, which resembles a spider's cob web. The topology comprises 2^R nodes for each ring, where R is the ring number which starts from 1. In the topology with R rings, there are $\Sigma 2^R$ total nodes [14]. For example, a RiCoBiT with three rings, each ring contains a different number of nodes. Ring 1 has 2 nodes, Ring 2 has 4 nodes, and Ring 3 has 8 nodes, which is depicted in Figure 9. The total number of nodes in the entire RiCoBiT is the sum of these values, which amounts to 14. The topology has a regular and symmetrical structure. It is organised, modular, and scalable, with the size only being constrained by fabrication technology.

Among the NoC topologies discussed earlier, mesh and torus are the most widely used. However, based on the analysis of performance parameters such as maximum hop count, wire length, and the number of wire segments required, it has been observed that RiCoBiT outperforms both of them.

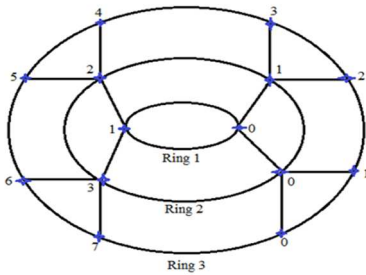


Fig. 11. Ring connected binary tree topology.

C. Flow Control Mechanisms for NoC

To develop an effective routing algorithm for a Network-on-Chip (NoC)[9][12][13][15], it is important for the designer to have knowledge of the various routing strategies and flow control mechanisms that are compatible with NoC architectures. These include store and forward routing, virtual cut-through routing, and wormhole routing, which are different network flow control mechanisms commonly used in NoCs.

Among the various flow control mechanisms [15] used in NoC architectures, the store-and-forward method is the most straightforward. This method involves temporarily storing data packets at each intermediate node before forwarding them to the next node, which requires a significant buffer capacity. However, this approach suffers from higher latency and the need for a large buffer size, which are considered its primary drawbacks [7][9].

To address the latency and buffer size limitations of the store-and-forward approach, a virtual cut-through [16] method has been developed for NoC architectures. This

alternative approach involves immediately transmitting packets upon arrival at a router, without the need for temporary storage. As a result, latency can be significantly reduced, even with a smaller buffer size, since packets are not held at intermediate nodes [15].

Worm-hole routing is a packet-switching technique that involves breaking data packets into small units called "flow control digits" or "flow control units" (flits). These flits are then transmitted through the network, with the path of transmission determined by the first flit. This means that subsequent flits follow the same path as the first flit that successfully makes it through the network.

Compared to other routing methods, worm-hole routing offers several advantages. It can reduce the limitations of other methods by enabling more efficient use of network resources, reducing congestion, and improving overall network performance. This is because worm-hole routing allows for parallel transmission of flits, enabling multiple packets to be transmitted simultaneously [7][8].

D. Routing Algorithms for NoC

Routing algorithms for Network-on-Chip (NoC) can be classified into two main categories: oblivious and adaptive. This categorization is depicted in the Figure 12.

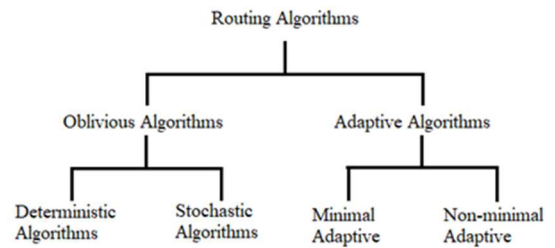


Fig. 12. Classification of routing algorithms.

Oblivious routing algorithms in NoC do not rely on any network condition or traffic information to route packets. There are two main types of oblivious routing algorithms: deterministic and stochastic [5][9][13]. Deterministic algorithms always follow a fixed route for packet transmission, whereas stochastic algorithms route packets randomly. The choice between these two types of algorithms depends on the specific requirements of the application and the desired trade-off between performance and complexity.

Minimal turn routing is a simple-to-implement type of oblivious routing algorithm used in NoC. This algorithm selects the direction for packet routing at each step, and examples of minimal turn routing include XY routing, pseudo-adaptive XY routing [8][9][13], and surrounding XY routing. In contrast, live lock-free routing algorithms such as west-first routing, north-last routing, and negative first routing guarantee deadlock avoidance by ensuring that packets do not get stuck in an infinite loop during routing. The selection of a specific routing algorithm depends on various factors, such as network topology, traffic patterns, and performance requirements.

Deterministic routing algorithms [8][9][14] are widely used in NoC, and some examples include shortest path routing, source routing, destination-tag routing, and topology adaptive routing. Shortest path routing finds the path with the minimum number of hops between source and destination nodes. In source routing, the source node determines the entire

path for a packet, and in destination-tag routing, each node appends a tag to the packet specifying the destination. Topology adaptive routing algorithms dynamically adjust the routing path based on network conditions and traffic patterns, enabling efficient packet transmission while avoiding congestion. The choice of a specific deterministic routing algorithm depends on various factors, such as network size, traffic patterns, and performance requirements.

Stochastic routing algorithms [7][8][10] in NoC are based on the assumption that every packet will eventually reach its destination. Unlike deterministic routing, stochastic routing does not follow a fixed path but instead randomly selects a path based on the network conditions and traffic patterns. This approach can lead to more efficient use of network resources, but it also introduces a certain level of uncertainty in the packet transmission process.

Adaptive routing algorithms [3][9][14] are an advanced version of oblivious routing in NoC, where routing decisions are made based on real-time network conditions and traffic patterns. The adaptive routing algorithms are preferred over oblivious routing in scenarios where network conditions are dynamic and constantly changing. Some examples of adaptive routing algorithms include minimal adaptive routing, fully adaptive routing, turn around routing, odd-even routing, and hot potato routing. Minimal adaptive routing selects the path with the minimum number of turns, while fully adaptive routing dynamically adjusts the routing path based on congestion levels and packet priority. Turn around routing and odd-even routing are based on the concept of deadlock avoidance, while hot potato routing aims to minimize the latency by forwarding packets to the nearest available node. The selection of a specific adaptive routing algorithm depends on various factors, such as network size, traffic patterns, and performance requirements.

In interconnection networks, oblivious and adaptive routing encounters significant issues such as deadlock, livelock, and starvation [5]. Deadlock occurs when several packets cannot move forward as they await the release of shared resources like packet buffers, ultimately leading to a network deadlock if a cycle of waiting packets forms. Livelock, on the other hand, arises when packets keep circulating through the network without reaching their destination, mainly in nonminimal routing algorithms. To avoid livelock, we can use two techniques: deterministic avoidance or probabilistic avoidance. The former involves adding a misroute count state to each packet to guarantee its progress, while the latter ensures that the probability of a packet remaining in the network for T cycles or less tends to zero as T tends to infinity. Moreover, in routing methods that are non-minimal and adaptive, age-based priority is a useful strategy to avoid livelock. However, different priorities can lead to a situation called starvation, where lower priority packets fail to reach their destinations as higher priority packets continuously reserve resources. To mitigate starvation, adopting a fair routing method or allocating some bandwidth solely for low-priority packets can be helpful.

III. COMPARATIVE ANALYSIS

In evaluating the performance of various topologies for NoC architectures, parameters such as maximum hop count, average hop count, number of wire segments, and wire length are commonly used [8][11][12].

A. Maximum Hop Count

The maximum hop count represents the longest distance among all the shortest paths in an array, which determines the latency of communication. As the maximum hop count increases, latency increases and performance decreases inversely.

B. Average Hop Count

The mean hop count is determined by averaging the hop counts across all pair combinations, providing a comprehensive measure of the system's overall performance.

C. Wire Segments Count

As the number of wire segments increases, the required routing area also increases. Consequently, this decrease in efficiency hampers the communication capability of packets within the network.

D. Wire length

Wirelength and the number of wire segments are essentially equivalent, as they both have a similar impact on network performance. Specifically, they exhibit an inverse relationship with the efficiency of packet traversal.

In such comparisons, the RiCoBiT topology has consistently shown superior performance relative to other topologies. The performance parameters of different topologies are quantitatively analyzed and presented in Tables I to IV. Table V shows the qualitative analysis of RiCoBiT, mesh and torus. This pointed out that RiCoBiT is performing better in terms of the parameters considered.

TABLE I. MAX HOP VS NUMBER OF NODES

Number of Nodes	Max Hop Count		
	Mesh Topology	Torus Topology	RiCoBiT Topology
4	2	2	1.17
8	3.65	2.82	2.64
16	6	4	4.33
32	9.31	5.65	6.17
64	14	8	8.08
128	20.62	11.3	10.08
256	30	16	12.08
512	43.25	22.62	14.08
1024	62	32	16.08
2048	88.5	45.25	18.08

TABLE II. AVERAGE HOP VS NUMBER OF NODES

Number of Nodes	Average Hop Count		
	Mesh Topology	Torus Topology	RiCoBiT Topology
4	1	1	0.5
8	2	1.5	0.75
16	2.5	1.75	1.75
32	4	3.5	3.25
64	5	4	4.25
128	8	6	6
256	10.5	7.5	7.25
512	15.5	11.5	9
1024	21	15.5	11
2048	31.5	24.25	12.5

TABLE III. WIRE-LENGTH VS NUMBER OF NODES

Number of Nodes	Wire Length		
	Mesh Topology	Torus Topology	RiCoBiT Topology
4	4	8	8
8	10.34	20.68	20
16	24	48	44

Number of Nodes	Wire Length		
	Mesh Topology	Torus Topology	RiCoBiT Topology
32	52.68	105.37	92
64	112	224	188
128	233.37	466.74	380
256	480	960	764
512	978.74	1957.49	1532
1024	1984	3968	3068
2048	3945.45	7890	6050

TABLE IV. NUMBER OF WIRE SEGMENTS VS NUMBER OF NODES

Number of Nodes	Number of Wire Segments		
	Mesh Topology	Torus Topology	RiCoBiT Topology
4	8	8	8
8	16	16	16
16	32	32	32
32	64	64	64
64	128	128	128
128	256	256	256
256	512	512	512
512	1024	1024	1024
1024	2048	2048	2048
2048	4096	4096	4096

TABLE V. QUALITATIVE ANALYSIS OF PERFORMANCE COMPARISON AMONG DIFFERENT TOPOLOGIES

Performance Parameters	Topologies		
	RiCoBiT	Mesh	Torus
Maximum Hop Count	Minumum	Maximum	Moderate
Average Hop Count	Minimum	Maximum	Moderate
Wire length	Moderate	Minimum	Maximum
Number of Wire Segments	Same	Same	Same

IV. OBSERVATION AND DISCUSSION

It is evident from the preceding discussion that the performance parameters of the commonly used NoC topologies deteriorate as the number of nodes increases. Based on a comprehensive analysis of Tables I to IV, it is evident that the RiCoBiT topology outperforms other topologies in the performance parameters max hop count and average hop count, particularly when dealing with large number of nodes.

V. CONCLUSION

The traditional bus architecture has several limitations that can be overcome by the adoption of NoC technology. NoC is the future of the semiconductor industry for designing multicore devices, and to take advantage of this technology, it is essential to understand its basic building blocks, including topology, routing, routing strategy, flow control mechanisms, and router microarchitecture. This paper provides a comprehensive discussion and comparison of different building blocks of NoC. Several topologies were analyzed, and RiCoBiT, Mesh, and Torus were found to be adaptable for further study based on various parameters such as the maximum hop count, number of wire segments required for communication, and wire length. This study revealed that for the future advancement of multicore processors, RiCoBiT emerges as a promising topology to be taken into consideration. The wormhole routing flow control mechanism is commonly used in NoC architectures. Additionally, adaptive routing algorithms are more commonly preferred for

NoC architectures since they can avoid deadlock compared to oblivious routing algorithms.

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